Inertial Relativity Mathematics

Tom DeGerlia

Category	**Relation**
Gravitational Time Dilation	
	$\frac{d\tau}{dt} = \sqrt{1-k}$
Raw Time Dilation	
	$\text{RTD} = \sqrt{k}$ $\text{RTD}^2 = k$
Space-Time Equivalence	
	$k = \left(\frac{I_1}{I_2}\right)^{\frac{1}{5}}$ $k = \left(\frac{\frac{2}{5}\, m_2\, r_2^2}{\frac{2}{5}\, m_1\, r_1^2}\right)^{\frac{1}{5}}$
Static Density (Isometric)	
	$k = \left(\frac{M_2}{M_1}\right)^{\frac{1}{3}}$ $M' = k^3 M$ $R' = kR$
Static Mass (Gravitational)	
	$k = \frac{R_S}{R}$ $R' = kR$ $R_S = kR$
Inertial Density	
	$k = \frac{I_2/V_2}{I_1/V_1}$
Mass–Radius Ratio	
	$k = \frac{M_2/R_2}{M_1/R_1}$ $k = \frac{M_2/R_2}{6.73295 \times 10^{26}}$
Static Radius	
Special Relativity	$k = \frac{v^2}{c^2}$
Isometric Form	$k = \left(\frac{m_2}{m_1}\right)^{\frac{1}{3}}$
Time Dilation Forms	
Normalized (vs. threshold)	$k = \frac{D}{D_{\text{crit}}} = \frac{M/R}{6.73295 \times 10^{26}}$
Raw (vs. other system)	$k = \frac{D_1}{D_2} = \frac{M_1 R_2}{M_2 R_1}$
GTD from normalized k	$\frac{d\tau}{dt} = \sqrt{1-k}$ (pace vs. flat spacetime)
GTD from raw k	$\sqrt{1-k}$ (pace vs. other system)
Recover D/D_{crit} from GTD	$\frac{D}{D_{\text{crit}}} = 1 - \text{GTD}^2$
DeGerlia Threshold	
	$D_{\text{crit}} = \frac{c^2}{2G} = 6.73295 \times 10^{26}$ kg/m $D_{\text{crit}} = \frac{M}{R_S}$ (for any mass)